HaDy_MZB

Habitat Dynamics MacroZooBenthos

A Python toolbox for analyzing patch-scale habitat dynamics

User manual

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GitHub repository:

https://github.com/HaDy-toolbox/HaDy_MZB

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Impressum

Authors of the toolbox and user manual

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Scientific background

This toolbox implements the framework presented in Lecrivain et al. (2026).

Technical support

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Toolbox testing

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1. Toolbox presentation

1.1 Purpose of the HaDy_MZB toolbox

Rivers are dynamic and habitat conditions change over time as discharge varies and interacts with channel morphology. These changes are especially relevant at the patch scale, where organisms experience hydraulic stress, habitat persistence, dewatering, and shifts between suitable and unsuitable habitat conditions.

HaDy_MZB is a free and open-source Python-based HaDy-toolbox for quantifying flow-driven habitat dynamics at the patch scale, with a focus on benthic macroinvertebrates — organisms particularly vulnerable to hydraulic stress due to their limited mobility and complex life cycles. However, the toolbox can also be adapted to other organism groups such as fish eggs and larvae, or vegetation, which may likewise be affected by limited mobility or exposure to changing habitat conditions.

HaDy_MZB links discharge time-series with precomputed hydraulic information for selected discharges to generate habitat time-series. The toolbox is designed to be flexible with respect to both the spatial and temporal scale of the analysis. The study reach can range from a few meters to tens of kilometers in length, while the temporal resolution of the discharge time-series can vary from minutes to hours. Based on these habitat time-series, the toolbox calculates four spatially explicit metrics that quantify habitat availability, habitat stability, and exposure to hydraulic stress over time (see Table 1). A patch represents a fixed spatial unit corresponding to the spatial and temporal perception of macroinvertebrates, typically about 0.25-0.50 m^2, defined by the computational mesh used in high-resolution hydrodynamic models (see more in Section 8).

The toolbox builds on the patch-scale approach developed by Bätz et al. (2025) and applied in Lecrivain et al. (2026). In particular, it puts into practice the logic of deriving habitat time-series from discharge scenarios and habitat distributions. Hydropeaking is a key application because frequent sub-daily discharge fluctuations strongly increase habitat dynamics. However, the toolbox can also be used more generally to analyze habitat responses to variable discharge regimes.

Table 1 Metrics derived by the HaDy_MZB toolbox

Metric	Definition	Relevance
M1 Habitat probability	Likelihood of different habitat conditions within a patch through time (Bätz et al.2025)	Time-integrated habitat availability; identifies dominant habitat types
M2 Habitat shifts	Daily number of times (i.e., frequency) that habitat conditions change to another suitable habitat type in a patch (Bätz et al.2025)	Indicates habitat stability and temporal persistence
M3 Drift risk	Risk of macroinvertebrate drift based on current velocity and up ramping rate experienced within each patch over time	Identifies exposure to critical passive drift conditions
M4 Desiccation risk	Risk of egg desiccation linked to hydropeaking-induced water level variations	Identifies exposure to critical desiccation duration

Who HaDy_MZB is for:

HaDy_MZB is designed for **researchers** and **practitioners** who want to assess discharge-driven habitat dynamics and ecological stressors over time in a spatially explicit way. It supports applications such as hydropeaking impact assessment, comparison of hydrological scenarios, evaluation of morphological complexity, and reach-scale mitigation planning.

The toolbox does not replace hydrodynamic modelling or ecological expert judgement. Instead, it provides a reproducible workflow for translating existing hydraulic or habitat data and discharge time-series into interpretable habitat-dynamics metrics.

1.2 Overview of the Workflow

The execution of the toolbox involves 3 steps (Figure 1):

- **Pre-processing:** the hydrodynamic simulation of the study area is carried out by the user. It provides spatially distributed data on water depth and current velocity for a representative range of discharges (input shapefile.shp). These data are then used to derive habitat types. The discharge time-series is required to describe the temporal evolution of discharge (input discharge.csv). See more in Section 3.
- **Processing**: the toolbox links discharge time-series with hydraulic data to generate habitat time-series and to calculate selected metrics.
- **Post-processing:** the resulting outputs can be visualized, mapped and further analyzed using GIS (see Section 5, and Section 9).

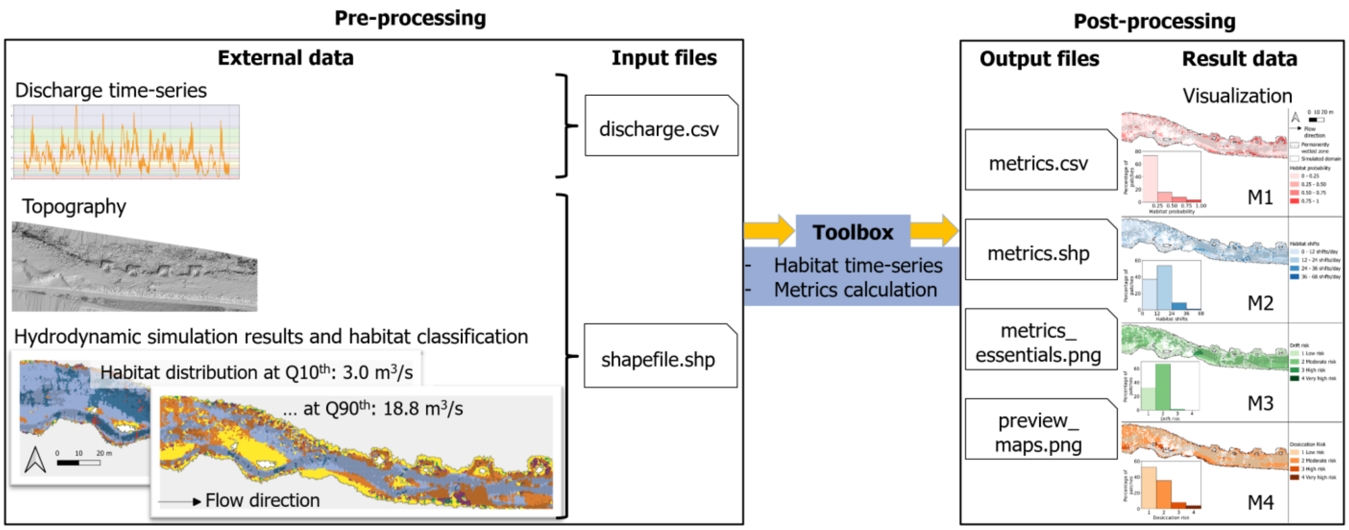


Figure 1 Overview of the execution procedure of the toolbox HaDy_MZB

The respective roles of the user and the toolbox are summarized in Table 2.

Table 2 Respective roles of the user and the toolbox throughout the workflow

Step	Responsibility	Activity
Pre-processing	User	Selects the discharge time-series and its timestep (time between two discharge values)
Pre-processing	User	Selects discharges of interest for the hydrodynamic simulation
Pre-processing	User	Selects the study area of interest
Pre-processing	User	Runs the hydrodynamic simulation
Pre-processing	User	Decides on a habitat classification method and relevant ecological thresholds
Processing	Toolbox	Builds **habitat time-series** from discharge time-series, water depth, current velocity and habitat classification method
Processing	Toolbox	**Calculates the selected patch scale metrics**
Processing	Toolbox	Exports results (i.e. metric values) in a shapefile, preview maps, and a csv document
Processing	Toolbox	Provides graphical plots
Processing	Toolbox	Provides additional functions for deeper analysis if needed
Post-processing	User	Analyses the output of the toolbox on GIS software
Post-processing	User	Adapts the toolbox to include additional plots or analyses if needed

1.3 Example case study

This manual follows a **running example dataset** provided with the toolbox (Windows operating system). Blue boxes such as the following provide practical guidance and shortcuts to quickly explore, reproduce, and adapt the workflow. Running the example case as an initial analysis is a good way to become familiar with the toolbox.

Example case study boxes are listed below:

- **Example Case Study Box 1 Quick** Input – Output example (no download or setup required)
- **Example Case Study Box 2:** From python repository setup to running the code
- **Example Case Study Box 3:** Outputs from the main.py file execution
- **Example Case Study Box 4** : Outputs from the plots.py file
- **Example Case Study Box 5 :** Outputs from the clustering.py file

Moreover, videos are accessible here to guide throughout the process, using the example case study:

- 0_HaDy_GitHub_general_introduction
- 1_HaDy_setup_and_first_run
- 2_HaDy_example_case_from_inputs_to_outputs
- 3_HaDy_example_case_QGIS
- 4_HaDy_local_yaml_for_tailored_analysis
- 5_HaDy_output_analysis
- 6_HaDy_advanced_focus_on_zone

Example case study Box 1: Quick Input – Output example (no download or setup required)

This box presents where to find the input and output files related to the example case study on GitHub (Figure 2). This does not require any installation, download or setup.

More information in the videos 0_HaDy_GitHub_general_introduction and 2_HaDy_example_case_from_inputs_to_outputs.

Example inputs (GitHub HaDy-toolbox following folders data > input):

- Discharge data (Week_timestep_10_min.csv)
- Input shapefile with water depth and current velocity value (Example_input_shapefile.shp)
- Habitat definition (see Section 4.3)

Example outputs (GitHub HaDy-toolbox following folders data > output > Example_output):

- Metrics (folder Metric_files)
- Maps (folder Metric_files > Preview_maps for the preview maps, and folder GIS_project containing the projects and the templates)
- Plots (folder Plots)
- Clustering (folder Clustering)

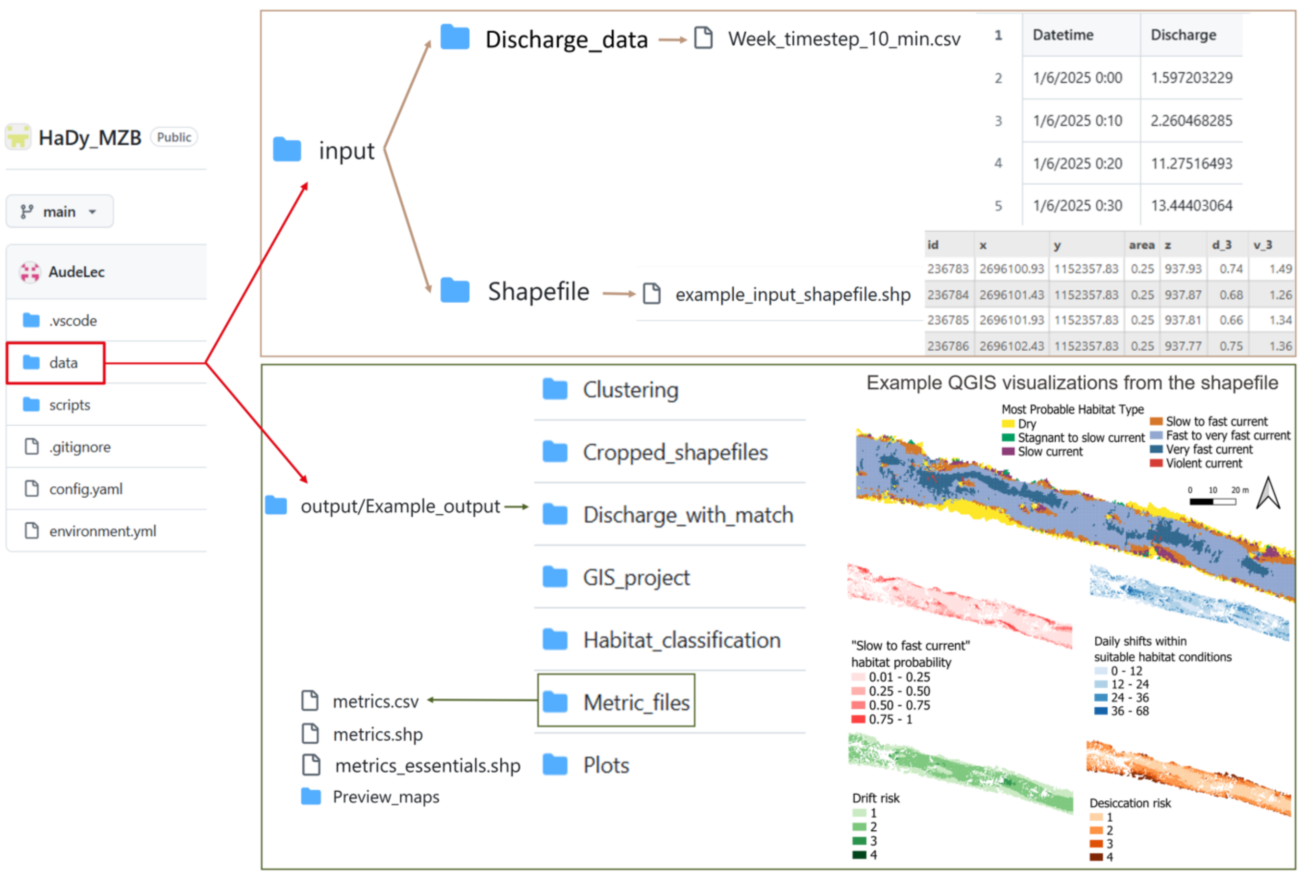


Figure 2 Quick input-output overview of the example case study on GitHub

2. Technical setup

This chapter explains how to install the toolbox and how to use it.

2.1 Prerequisites

If the following usual **python core tools** and **plugins** mentioned below are not installed on the laptop, check Section 10.1.

Required tools:

- Python
- Git
- Anaconda/Miniconda
- Visual Studio Code (recommended), **including the python extension**
- Python extension for VS Code

2.2 GitHub repository setup

All steps are explained and illustrated in Example Case Study Box 2 and in the [video 1_HaDy_setup_and_first_run].

Step 1: Create a working directory where the Git repository will be cloned

Create a folder where the Git repository will be cloned.
The folder can be located anywhere on the system, provided that its path does not contain any spaces.
The folder can be given any name, as long as the name does not contain spaces.

Step 2: Clone the repository

Open a terminal and go to the created working directory: enter the command "cd" and the absolute path to your directory (**cd path/to/your/directory**).

Once in the working directory, paste the cloning command (Git needs to be installed otherwise this will not work, see Section 2.1):
git clone https://github.com/HaDy-toolbox/HaDy_MZB.git

Step 3: Setup the virtual environment

In the terminal (for macOS and Linux), or anaconda prompt (Windows) go to **the cloned folder called HaDy_MZB** in the working directory, and paste the following commands:
conda env create -f environment.yml, and then **conda activate Venv_HaDy_MZB**

Step 4: Open the project on VS code and select the python interpreter

Open the software VS code previously installed, open the cloned HyDy_MZB folder.

To select the interpreter: Ctrl+Shift+P (to get the command panel) → type "Python: Select Interpreter" → Choose Venv_HaDy_MZB.

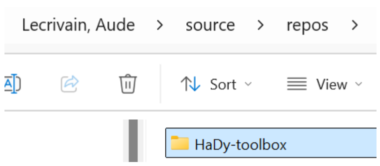
Step 5: Run the code

Go to the python file *main.py* and click the play button (see).
The same steps can be taken in the *plots.py* or *clustering.py* for further analysis (see Section 4.6).

ATTENTION	If the analysis is tailored, make sure that the local.yaml file has been adapted **and saved** (more in Section 3.2) before running the codes. Otherwise, the outputs will not correspond to the specified parameters.

Example case study Box 2: *From python repository setup to running the code*

Step 1: Create a working directory where the Git repository will be cloned



Step 2: Clone the repository

In the terminal, type in "cd" and paste the absolute path to your directory (no spaces allowed)

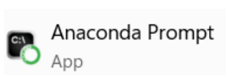
```
PS C:\Users\lecrivau> cd C:\Users\lecrivau\source\repos\HaDy-toolbox
```

Then type in the cloning command

```
PS C:\Users\lecrivau\source\repos\HaDy-toolbox> git clone https://github.com/HaDy-toolbox/HaDy_MZB.git
```

Step 3: Setup the virtual environment

On windows, use Anaconda prompt



Go to the cloned HaDy_MZB and type in the commands

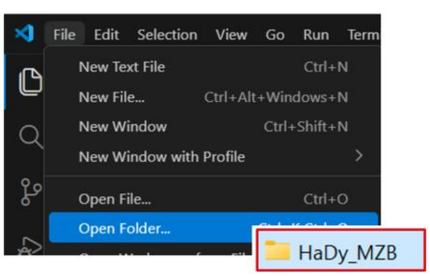
```
C:\Users\lecrivau\source\repos\HaDy-toolbox>cd C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB

C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB>conda env create -f environment.yml

rs\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB>conda env create -f environment.yml
```

You now see that the environment is activated. `(Venv_HaDy_MZB) C:\Users\lecrivau>`

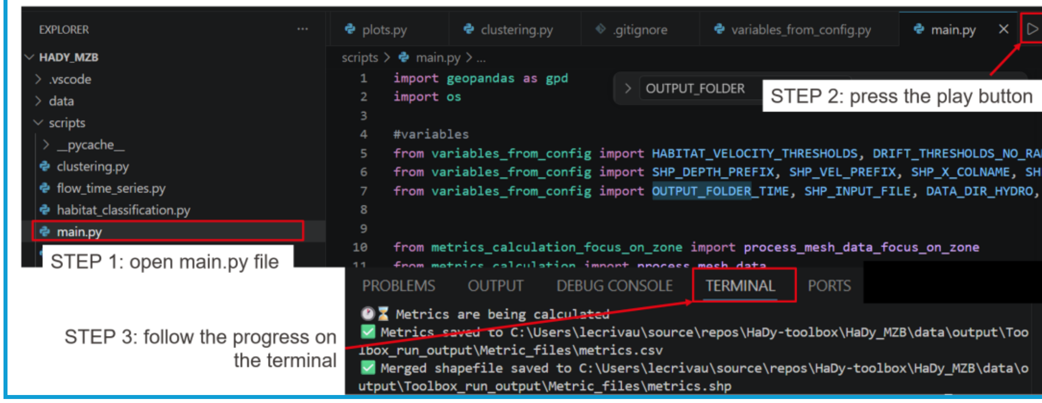
Step 4: Open the project on VS code and select the python interpreter



To select the interpreter: Ctrl+Shift+P (to get the command panel) → type "Python: Select Interpreter" → Choose `Python 3.11.11 (Venv_HaDy_MZB) ~\.conda\envs\Venv_HaDy_MZB\python.exe`

Step 5: Run the code

For the example case study, run the code without modifying the configuration file (config.yaml)



2.3 For later use: get the latest version from GitHub

Since the toolbox is in constant evolution, make sure to use the latest version available. To update it, “pull” the changes from the GitHub folder as done on Figure 3.

This can be done directly in the Visual Studio code terminal as follows: command **git pull origin main**

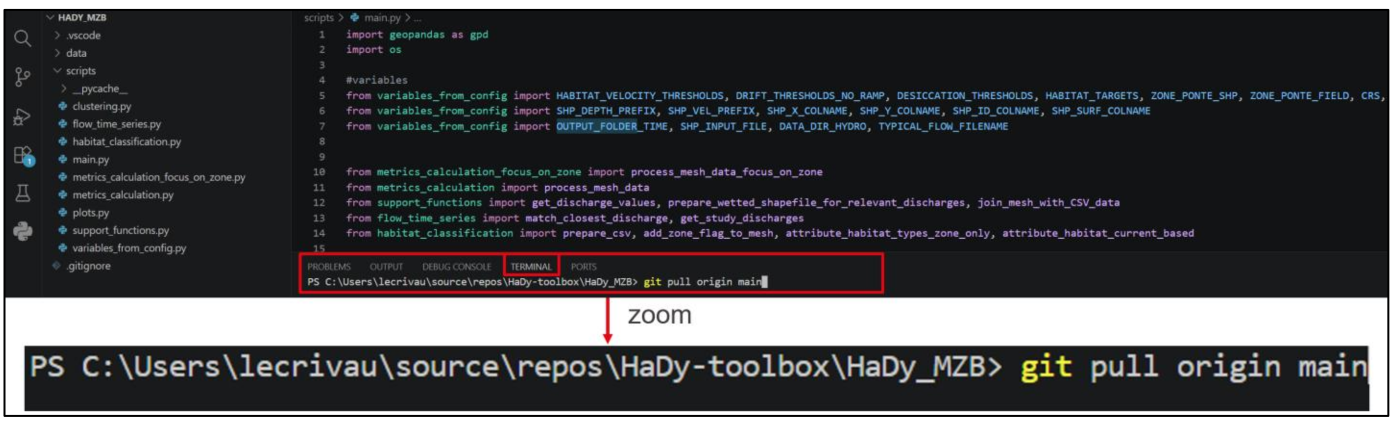


Figure 3 Example of how to pull the last version

ATTENTION

Running this command replaces the contents of the scripts and input_data/local folders with the latest files from the GitHub repository. Be sure to back up any local changes before executing the command, as they will be overwritten.

3. Input and configuration

3.1 Input files

Input files are located in the data > input folder, which contains two subfolders: Discharge_data and Shapefile. The relevant files should be placed in the appropriate folder to be included in the analysis. The following sections describe the required file contents and minimum requirements.

Discharge time-series input data

Minimum requirements:

- A CSV file containing discharge variations over time (m^3/s), as illustrated in Figure 4.
- The file must be located in data > input > Discharge_data.
- The column containing discharge values must be named Discharge (exactly as written, including the capital letter). The format of the other columns is not relevant, and they may be retained or deleted (e.g., Figure 4, right). A date column is included for user convenience but is not read by the code. The timestep between consecutive discharge values is instead specified by the user in the configuration file.
- The timestep specified in the configuration file must match the timestep between consecutive rows in the input file (see Section 3.2). The toolbox requires this information, for example, to calculate daily metrics. It is therefore the user's responsibility to provide an input file with the appropriate timestep for the intended analysis.
- Discharge values must be formatted as numbers, with a dot (.) used as the decimal separator.

ATTENTION

Common issues:

- **Incorrect file location:** The input file must be placed in the specified folder.
- **Incorrect file format:** The input file must be provided in CSV format, not as an Excel file.
- **Incorrect Discharge column name:** The discharge column must be named Discharge exactly, including the capital letter.
- **Incorrect decimal separator:** Decimal values must use a dot (.), not a comma (,).
- **Missing values:** Missing values must be addressed before running the analysis.
- **Inconsistent timesteps:** The timestep specified in the configuration file must match the timestep between consecutive discharge values in the input file. An incorrect timestep can lead to incorrect metric values.

Date (TU)	Valeur (en mÂ³/s)	Statut	Qualification	MÃ©thode	ContinuitÃ©
2025-06-24T00:00:00.000Z	14.4	12	20	8	0
2025-06-24T00:10:00.000Z	14.4	12	20	8	0
2025-06-24T00:20:00.000Z	14.4	12	20	8	0
2025-06-24T00:30:00.000Z	14.4	12	20	8	0

Date (TU)	Discharge
2025-06-24T00:00:00.000Z	14.4
2025-06-24T00:10:00.000Z	14.4
2025-06-24T00:20:00.000Z	14.4
2025-06-24T00:30:00.000Z	14.4

Figure 4 Example of data downloaded from the HydroPortail platform (left) and of the data retained and renamed for the analysis (right). The Date column is retained here for user convenience but is not read by the code and is therefore not mandatory.

Shapefile input data

Minimum requirements:

- A polygon shapefile (.shp and associated files) containing water depth and current velocity data for the discharges of interest (e.g., Figure 5). These data should originate from hydrodynamic simulations.
- The shapefile must be located in data > input > Shapefile.
- The prefixes of the relevant column names must be specified in the configuration file (see Section 3.2).
- The shapefile must contain the data listed in Table 3.

Table 3 Data that needs to be in the input polygon shapefile

Id	Id of the patch - One row per patch
x	Coordinates of the patch center
y	
Area m2	Area associated with each patch, used to normalize metrics by area
Water depth for each discharge **(meters)**	Water depth values for each discharge of interest. Decimals separated by "_". For example, with the prefix d_, discharges of 31.8 and 6 m^3/s correspond to d_31_8 and d_6, respectively.
Water current velocity for each discharge **(meters/second)**	Current velocity values for each discharge of interest. Decimals separated by "_". For example, with the prefix v_, discharges of 31.8 and 6 m^3/s correspond to v_31_8 and v_6, respectively.

id	x	y	area	z	d_1	v_1	d_2	v_2	d_3	v_3	d_4_2	v_4_2	d_5_5	v_5_5	d_7_4	v_7_4	d_9	v_9	d_10_6	v_10_6	d_12	v_12
276139	2696154.44	1152347.31	0.25	937.3	0.33	0.62	0.43	0.88	0.53	1.18	0.58	1.29	0.65	1.43	0.74	1.57	0.8	1.65	0.87	1.71	0.92	1.76
276140	2696154.94	1152347.31	0.25	937.3	0.34	0.6	0.45	0.87	0.55	1.17	0.59	1.28	0.66	1.42	0.75	1.56	0.82	1.65	0.88	1.71	0.93	1.76
276141	2696155.44	1152347.31	0.25	937.28	0.35	0.57	0.46	0.82	0.56	1.11	0.61	1.21	0.68	1.36	0.77	1.5	0.84	1.59	0.9	1.65	0.95	1.7

Figure 5 Example of data in the polygon shapefile

ATTENTION

Common issues:

- **Missing files:** If only the .shp file is present in data > input > Shapefile, the toolbox will not run. All associated shapefile components are required (see the files provided in the Example_input_shapefile folder).
- **Empty or N/A values:** Empty or N/A values must be replaced with 0 for the toolbox to run the analysis.
- **Data consistency:** If a patch is dry under a given discharge (i.e., water depth = 0), the corresponding current velocity must also be set to 0.
- **Missing data:** Each patch must have both current velocity and water depth data for every discharge included in the analysis.
- **Incorrect formatting:** The water depth and current velocity field names must include the corresponding discharge value. Decimal points in discharge values must be replaced by an underscore (e.g., d_31_8 and v_31_8 for 31.8 m^3/s, or

d_6 and v_6 for 6 m^3/s). The prefixes used for water depth columns must be consistent across all water depth fields, and the prefixes used for current velocity columns must be consistent across all current velocity fields. These prefixes must also match those specified in the configuration file (see Section 3.2).

- **Spatial representation:** The input data must be provided as a polygon shapefile, rather than a point shapefile, to ensure a continuous spatial representation of the results.

3.2 Configuration

Minimum requirements:

- To run an analysis that differs from the example case study, a local.yaml file must be created in the repository. This file should be created by copying config.yaml and making the necessary modifications to tailor the analysis (see the video 4_HaDy_local_yaml_for_tailored_analysis).
- The local.yaml file must be located in the root of the repository, in the same location as config.yaml.

Common issues:

- **Incorrect file name:** The file must be named local.yaml exactly. It is the user’s responsibility to create this file when tailoring the analysis parameters.
- **Missing local.yaml file:** If no local.yaml file is present, the toolbox uses config.yaml by default, which corresponds to the example case study.
- **Analysis does not match the specified parameters:** The local.yaml file must be saved after any modifications. Otherwise, the toolbox may run using the previously saved parameters.
- **Previous analysis overwritten:** Each new toolbox run stores its outputs in data > outputs > Toolbox_run_output, overwriting any files already present in this folder. Analyses that need to be retained should therefore be copied to another folder or renamed before running the toolbox again.

Table 4 describes the variables that need to be adapted by the user.

Table 4 Detail of the configuration variables

Input files

Name of the variable	Signification	Status
input_shp_filename	Name of the input shapefile located in the data > input > Shapefile folder	Mandatory
crs	CRS of the GIS project	Mandatory
input_discharge_timeseries_ filename	Name of the discharge time-series input file located in the data > input > Discharge_data folder	Mandatory
time_step_min	Timestep of the discharge time-series in minutes (float, e.g., 10.0 for a 10-minute timestep). /!\ Must match the timestep of the discharge time-series input file	Mandatory
shp_id_colname	Name of the id column identifying each patch in the input shapefile	Mandatory
shp_x_colname	Name of the column containing the x-coordinate of each patch in the input shapefile.	Mandatory
shp_y_colname	Name of the column containing the y-coordinate of each patch in the input shapefile.	Mandatory
shp_area_colname	Name of the column containing the area of each patch, used for potential area-based metric normalization.	Mandatory
shp_depth_prefix	Prefix preceding the discharge values in the input shapefile (e.g., d_ in d_30_2).	Mandatory
shp_vel_prefix	Prefix preceding the discharge values in the input shapefile (e.g., v_ in v_30_2).	Mandatory

Habitat classification thresholds

Name of the variable	Signification	Status
depth_threshold	Minimum water depth (m) required for a patch to be considered wet (float, e.g., 0.01 for 1 cm). Patches are considered wet when their water depth is greater than this threshold.	Mandatory
habitat_velocity_thresholds	Habitat classification thresholds based on current velocity and water depth (see Section 4.3 Classify habitats)	Mandatory
number_of_habitats	Number of habitat types defined by the habitat classification. For example, a classification ranging from 0 (dry) to 6 (highest velocity class) contains 7 habitat types.	Mandatory

ATTENTION

Common issues:

- **Incorrect number of habitats:** Ensure that the number of habitat types is counted correctly. For example, habitat classes ranging from 0 to 6 correspond to 7 habitat types. An incorrect number of habitat types will prevent the plots from being generated correctly.
- **Changes to habitat classification:** If the habitat classification method is modified, the attribute_habitat_current_based function in habitat_classification.py must also be adapted accordingly.

Metrics calculation

Name of the variable	Signification	Status
target_habitat	Target habitat type for which metrics are calculated. Metrics are only calculated for patches that experience this habitat type at least once during the habitat time series. The value must be an integer corresponding to one of the habitat types defined in the habitat classification. To obtain results for different target habitats, the analysis must be run separately for each target habitat.	Mandatory
metrics_to_compute	Selection of the metrics to calculate. Set to true to calculate a metric and false otherwise.	Mandatory
shift_all_daily	Daily shifts between all habitat types.	Mandatory
shift_targ_daily	Daily shifts between the habitat types defined as suitable in suitable_hab_shifts.	Mandatory
shift_dry_daily	Daily shifts to dry conditions, defined as water depth below depth_threshold.	Mandatory
dry_max	Maximum consecutive duration of dry conditions, expressed as a number of timesteps.	Mandatory
desiccation_risk	Desiccation risk calculated using the thresholds defined in desiccation_thresholds (see Section 4.4)	Mandatory
drift_percentile	Drift risk based on the specified drift percentile (see Section 4.4 Metrics calculation)	Mandatory
drift_max	Maximum associated drift risk value (see Section 4.4 Metrics calculation)	Mandatory
drift_durations	Number of timesteps spent in each drift risk category.	Mandatory
suitable_hab_shifts	List of habitat types considered suitable. Shifts between these habitat types are used to calculate shift_targ_daily.	Mandatory
start_at_first_occurrence	Set to true to start the desiccation metric calculation after the first occurrence of the target habitat in the habitat time series. Set to false to start the calculation at the beginning of the time series.	Mandatory

desiccation_thresholds	Desiccation thresholds based on the consecutive duration spent outside the water, expressed in hours (see Section 4.4).	Mandatory
up_ramp	Set to true to use drift_thresholds_with_ramp for drift calculations, taking the up-ramping rate into account. Set to false to use drift_thresholds, without considering the ramp rate.	Mandatory
drift_thresholds_with_ramp	Drift thresholds based on current velocity thresholds (m/s), taking the up-ramping rate into account. Used when up_ramp is set to true (see Section 4.4).	Conditional on up_ramp
drift_thresholds	Drift thresholds based on current velocity thresholds (m/s), without considering the up-ramping rate. Used when up_ramp is set to false (see Section 4.4Metrics calculation).	

ATTENTION

The user is responsible for determining desiccation and drift threshold values appropriate for the macroinvertebrate taxa and habitat type considered in the study. The user must also determine whether an up-ramping threshold is relevant for the calculation of drift risk.

Optional: advanced parameters for a focus on a specific zone

Name of the variable	Signification	Status
focus_on_zone	Set to true to focus the analysis on a specific zone (e.g., gravel bars for desiccation of eggs) and false otherwise.	Mandatory
zone_interest_filename	Name of the shapefile defining the zone of interest, including its associated files, located in the data > input > Focus_zone folder.	Conditional on focus_on_zone
id_polygon_zone_interest	Name of the ID column identifying the polygons in the zone of interest (e.g., individual gravel bars), allowing them to be compared. This column must have a different name from the shp_id_colname column in the input shapefile.	
zone_interest_field	Name assigned to the zone of interest in the output files. Patches belonging to the zone are assigned a value of 1, while other patches are assigned a value of 0.	

Optional: files paths for plots and clustering (further analysis)

Name of the variable	Signification	Status
base_output_path	Output folder located in data > output, containing the data to be used for further analysis.	Optional

final_csv_path	Relative path to the output CSV file containing the calculated metric values.	Optional
final_shp_path	Relative path to the output shapefile containing the calculated metric values.	Optional
static_habitat_csv_path	Relative path to the CSV file containing the habitat classification of each patch.	Optional

Tailoring an analysis

It is recommended to first run the example case study to become familiar with the toolbox and its workflow.

Once the example case study has been successfully run, the analysis can be progressively tailored.

ATTENTION	Whenever a parameter or code is modified, save the corresponding file before running the code for the changes to be taken into account.

As a first step, the following simple modifications can be tested:

Create a local.yaml file by copying config.yaml (see video 4_HaDy_local_yaml_for_tailored_analysis) and modify one parameter at a time.

Simple modifications:

- Change the target habitat
- Modify one habitat classification threshold
- Replace the discharge time-series and adapt the timestep accordingly
- Activate or deactivate one metric

More advanced modifications:

- Change desiccation thresholds
- Change drift thresholds
- Run *plots.py* or *clustering.py*
- Change the habitat classification. In this case, adapt both local.yaml and the attribute_habitat_current_based function in habitat_classification.py accordingly

Changing the spatial focus of the analysis:

- Set focus_on_zone to true
- Add a focus zone shapefile and complete the corresponding variables in local.yaml: zone_interest_filename, id_polygon_zone_interest, and zone_interest_field
- Adapt the habitat classification method to the specific scope of the analysis, if necessary

Running a fully tailored analysis:

Once familiar with the toolbox, the analysis can be adapted to specific study requirements. The following workflow should be followed:

1. Add the input files to the appropriate input folders, as described for the example case study.
2. Create local.yaml by copying config.yaml, then adapt the parameters to the requirements of the analysis and save the file (Ctrl+S). See the video 4_HaDy_local_yaml_for_tailored_analysis.
3. Adapt the habitat classification, if necessary. If the habitat classification method has been modified, update the relevant function in habitat_classification.py accordingly and save the changes (Ctrl+S).
4. Run main.py to perform the analysis.
5. Adapt plots.py if necessary to account for new habitat types, including habitat names and colors. Save the changes (Ctrl+S) before running the script.
6. Run clustering.py, if clustering is part of the analysis.
7. Explore the results in a GIS software. Import the output shapefile and, when using QGIS, apply the default layouts provided with the toolbox. See the video 3_HaDy_example_case_QGIS.

4. Full workflow explanation

This chapter explains what happens inside *main.py*, from input data to final metrics.

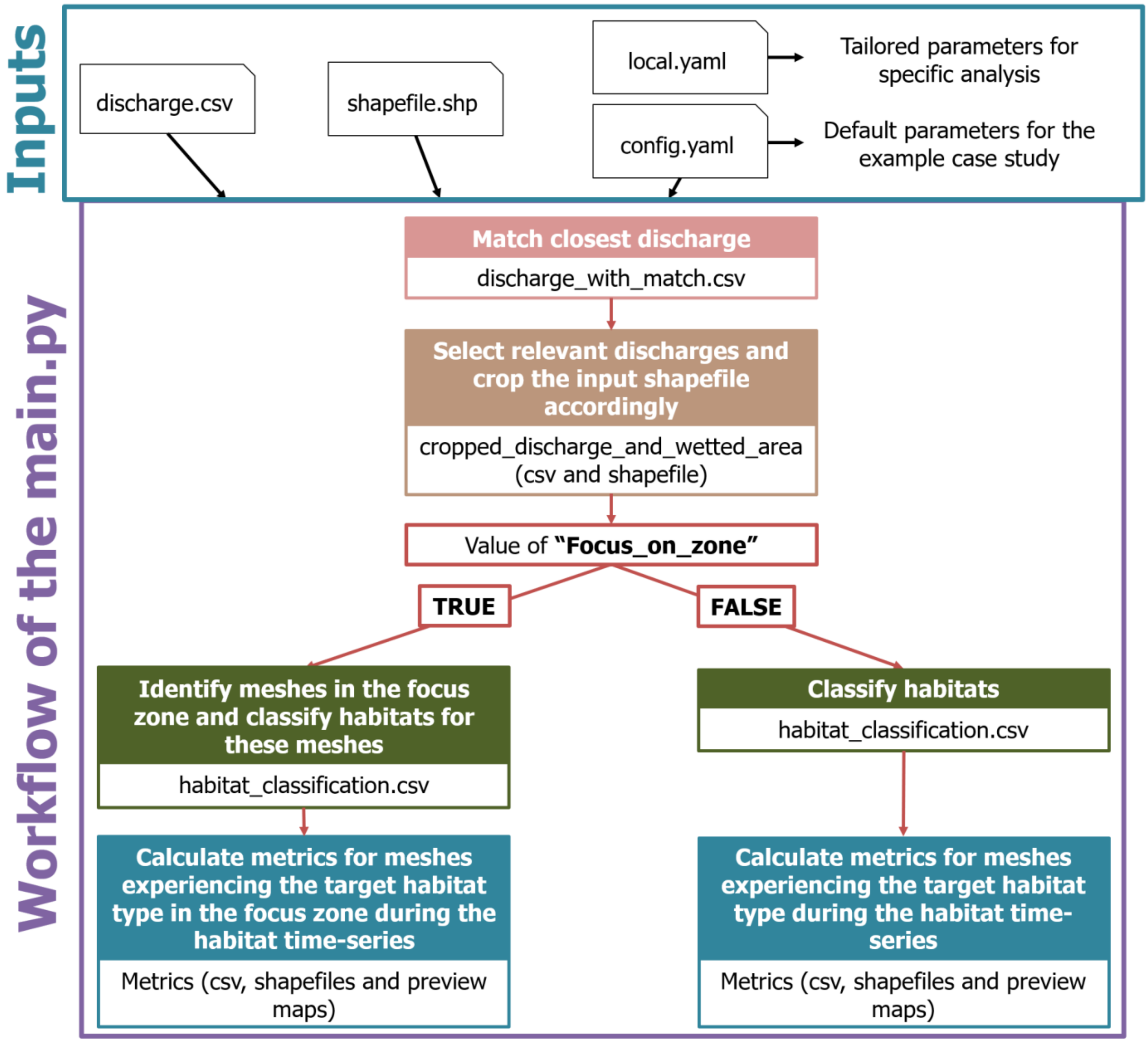


Figure 6 Full workflow of the main.py file

4.1 Match discharge time-series to the closest simulated discharges

To run the hydrodynamic simulations, a set of discharges of interest must first be selected. In the example case study, these discharges cover the 1st to 99th percentile range of the discharge time series, following the approach used in Lecrivain et al. (2026). However, this is only an example, and other methods can be used to select the discharges of interest.

The selected discharges should be representative of the discharge time series. For each of these discharges, the hydrodynamic simulations must provide water depth and current velocity values for every patch in the study domain, as described for the input shapefile in Section 3.1.

The toolbox then assigns each discharge in the input discharge time series to the closest discharge for which hydrodynamic data are available. The corresponding discharge is recorded in the Corresponding_known_discharge column, as illustrated in Figure 7.

No interpolation is performed for discharges for which hydrodynamic data are not available. If greater precision is required, or if more extreme discharges need to be considered, the input hydrodynamic data must be adapted accordingly. For example, additional hydrodynamic simulations can be performed for more extreme discharges.

Datetime	Discharge	Corresponding_known_discharge
1/6/2025 0:00	1.597203229	2
1/6/2025 0:10	2.260468285	2
1/6/2025 0:20	11.27516493	10.6
1/6/2025 0:30	13.44403064	13.4
1/6/2025 0:40	11.49950344	12
1/6/2025 0:50	9.056749663	9

Figure 7 Example of discharge time-series where each discharge is associated to the closest known one from the input shapefile provided by the user

4.2 Select relevant discharges and crop the input shapefile

Based on the Corresponding_known_discharge values, the toolbox determines the list of discharges to retain for the study. To reduce computational time, the input data are then cropped to retain:

- Only data corresponding to discharges within the scope of the study.
- Only patches that are wet under the maximum discharge considered.

4.3 Classify habitats

The user is responsible for defining the habitat classification method according to the scope of the analysis. Figure 8 presents the water depth- and current velocity-based classification used in the example case study. Similar approaches can be implemented with different numbers of habitat types and different threshold values. These parameters must be defined and adapted in both the configuration file (local.yaml) and the habitat_classification.py file.

Color	Current Velocity Class
	Dry
	Stagnant to slow current $v_{40} < 5$ cm/s
	Slow current $5 \leq v_{40} < 25$ cm/s
	Slow to fast current $25 \leq v_{40} < 75$ cm/s
	Fast to very fast current $75 \leq v_{40} < 150$ cm/s
	Very fast current $150 \leq v_{40} < 250$ cm/s
	Violent current $v_{40} \geq 250$ cm/s

Figure 8 Example of macroinvertebrate habitat classification at community-level (adapted from Tonolla 2023)

Another example focuses on a specific zone, with focus_on_zone set to true. This approach is implemented in the attribute_habitat_types_zone_only function in the habitat_classification.py script and targets egg deposition on gravel banks. Habitat type -1 indicates that a patch is located outside the zone of interest (gravel banks in this example), while the other habitat types are illustrated in Figure 9.

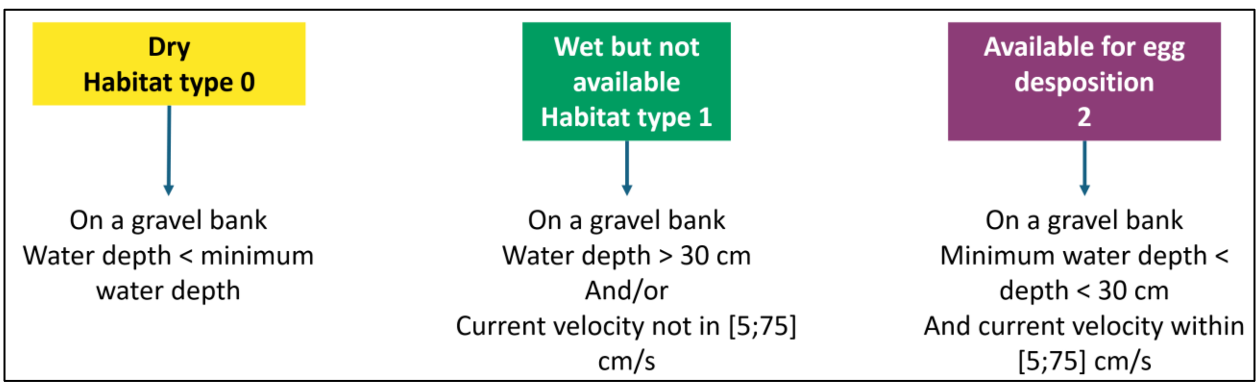


Figure 9 Example of habitat classification focusing on egg desiccation on gravel banks

ATTENTION

When modifying code in the repository (e.g., habitat_classification.py), make sure to save the changes before running the code for them to be taken into account. In the file editor, this can be done using Ctrl+S.

As previously mentioned, modified files should also be saved in a safe location before pulling the latest version of the Git repository. Pulling the latest version may overwrite local changes.

4.4 Metrics calculation

Scope of the metrics calculation

The simulated domain includes all patches that are wet under the maximum discharge of the discharge time series. Habitat probabilities are calculated for all these wetted patches.

Metrics are calculated only for patches that experience the target habitat type at least once during the habitat time series. The metrics to be calculated can be selected by the user in the configuration file.

Description of the metrics

The four patch-scale macroinvertebrate habitat metrics presented in Table 5 Overview of the four patch-scale metrics used to characterize macroinvertebrate habitat dynamics, including definitions, calculation are derived from the habitat time series and quantify complementary ecological aspects of habitat dynamics. Habitat probability (M1) and habitat shifts (M2) describe the availability and temporal persistence of suitable conditions and are therefore generally related to habitat quality. Drift (M3) and desiccation (M4) describe exposure to potentially harmful hydraulic events and their associated ecological risks.

Table 5 Overview of the four patch-scale metrics used to characterize macroinvertebrate habitat dynamics, including definitions, calculations, ranges, and ecological relevance

Metric	Definition	Calculation	Range	Relevance
M1) Habitat probability	Likelihood of experiencing different habitat conditions within a patch over time (Bätz et al. 2025).	$M1 = \frac{\sum t_H}{T}$	0 → 1 Min Max	Time-integrated habitat availability; identifies dominant habitat types
M2) Habitat shifts	Daily number of habitat changes to another suitable habitat type within a patch (Bätz et al. 2025).	$M2 = \frac{\sum S}{T}$	0 → Max. time steps Min Max	Indicates habitat stability and temporal persistence
M3) Drift risk	Risk of macroinvertebrate drift based on current velocity and, where relevant, the up-ramping rate experienced within each patch over time.	$M3 = P_{90}$ (Rdrift)	1 → 4 Min Max	Identifies exposure to critical passive drift conditions
M4) Desiccation risk	Risk of egg desiccation associated with hydropeaking-induced water-level variations.	M4 = max (Rdes)	1 → 4 Min Max	Identifies exposure to critical desiccation duration

t_H = duration of occurrence of a specific habitat type over the time-series
T = total duration of the time-series
S = habitat-shift between two consecutive time steps
P_{90} = 90th percentile value over the time-series

max = maximum value over the time-series
Rdrift = drift risk as defined in Table 6
Rdes = desiccation risk as defined in Table 7

Drift and desiccation risks are based on threshold values that depend not only on the taxa considered, but also on the available data and the habitat type targeted by the analysis.

Table 6 and Table 7 present the threshold values used in the example case study.

Table 6 focuses on drift risk for two habitat types: "Slow" (habitat type 2 in the toolbox) and "Slow to intermediate" (habitat type 3 in the toolbox). For the "Slow" habitat type, the analysis considers the up-ramping rate (up_ramp set to true in the configuration file, with a corresponding value specified in the drift ramp thresholds). The "Slow to intermediate" habitat type does not consider the up-ramping rate.

The parameters specified in the configuration file determine which drift calculation function is used by the metric calculation scripts. It is the user's responsibility to determine whether the up-ramping rate should be considered, based on the available data and the scope of the study.

Table 6 Macroinvertebrate drift risk thresholds and corresponding metric values for the two habitat types considered in the example case study.

Metric value (R_{drift})	**Drift risk class**	**Habitat type**		
		Slow [a]		**Slow to intermediate [b]**
		Current velocity threshold	**Up-ramping rate**	**Current velocity threshold**
1	Low	$v_{40} \leq 0.4$ m/s	Not relevant	$v_{40} \leq 1.2$ m/s
2	Moderate	$0.4 < v_{40} < 0.7$ m/s	Not relevant	$1.2 < v_{40} < 2.1$ m/s
		$0.7 \leq v_{40} \leq 1$ m/s	< 1 cm/min	
3	High	$0.7 \leq v_{40} \leq 1$ m/s	≥ 1 cm/min	$2.1 \leq v_{40} \leq 3$ m/s
4	Very high	$v_{40} > 1$ m/s	Not relevant	$v_{40} > 3$ m/s

[a]: thresholds taken from the flume experiments in Tonolla et al. (2019) and incorporating findings from Friese et al. (2025).
[b]: thresholds mathematically extrapolated from the “slow” current habitat, by proportionally adjusting the upper limit of the typical current velocity range. Up-ramping rate was not included to avoid introducing additional assumptions.

Table 7 presents the example of egg desiccation risk threshold values implemented in the toolbox.

Table 7 Macroinvertebrate desiccation risk thresholds and metric values based on Kennedy et al. (2016) and Jaulin (2025)

Metric value (R_{des})	**Desiccation risk class**	**Max. dry duration**
1	Low	≤ 1 hour
2	Moderate	1 - 5 hours
3	High	5 - 9 hours
4	Very high	> 9 hours

The start_at_first_occurrence parameter can affect the calculated desiccation risk. If set to true, desiccation risk is calculated only after the first occurrence of the target habitat type. If set to false, the calculation starts at the beginning of the discharge time series.

For example, for an analysis focusing on egg desiccation on gravel banks, setting start_at_first_occurrence to true means that the desiccation risk calculation starts after the first occurrence of the target habitat type considered suitable for egg deposition.

Advanced statistics when focusing on a certain zone

When focus_on_zone is set to true, additional statistics, such as median and quartile values, are calculated for the target habitat type and dry conditions. These statistics provide a more detailed understanding of habitat probability distributions (see Table 11 in Section 5.1 and the corresponding part of Section 5.2).

4.5 Estimation of the computational time

Computational time can vary substantially depending on the spatial and temporal extent of the analysis. Examples of estimated computational times are provided inTable 8Figure 7.

Table 8 Estimation of computational times based on the number of patches and of timesteps included in the analysis

Total number of patches	Number of patches for metrics calculation	Number of timesteps	Estimated computational time
30 446	25 769	1008	~ 5 minutes
365 000	27 350	1008	~ 50 minutes
664 000	76 060	4393	~ 4 hours

4.6 Additional functions

Table 9 Association between the Python script to run and the corresponding outputs

Output to get	Python file to run
Metrics results and preview maps	*main.py*
Plots	*plots.py*
Clustering	*clustering.py*

Before running the additional scripts, the paths to the files to be analyzed (base_output_path, final_csv_path, final_shp_path, and static_habitat_csv_path) must be specified in the configuration file (see Section 3.2). The configuration file must then be saved before running the scripts.

The metrics used by the plotting and clustering functions are determined from the configuration file. The selected metrics must therefore match the metrics available in the files being analyzed. For example, if desiccation_risk is set to true in the configuration file, the corresponding desiccation_risk field must also be present in the input file, as the functions will search for this field.

Plots

Output location: data > outputs > Toolbox_run_output > Plots

The plots focus on two geographical extents:

- **The entire simulated domain:** habitat probabilities and the most probable habitat types are assessed for all simulated patches. This provides an overview of the spatial distribution of habitats and helps explain why some patches do not experience the target habitat type during the habitat time series.
- **The metric calculation domain:** only patches that experience the target habitat type at least once during the habitat time series are included.

If the habitat classification is modified, the habitat information used by the plotting functions must also be adapted accordingly. In particular, the habitat labels and colors defined in plots.py need to correspond to the habitat codes used in the analysis. The appropriate label dictionary also needs to be adapted depending on whether focus_on_zone is set to true or false. If the habitat codes in the analysis do not match the codes defined in these dictionaries, the plotting functions may fail to generate the plots correctly.

Additional or more targeted plots can be created by modifying existing functions or adding new ones. As previously mentioned, pulling the latest version of the repository from GitHub can overwrite local modifications. Any customized code should therefore be saved in a separate, safe location before updating the repository.

Examples of outputs are presented in Section 5.2.

Clustering

Output location: data > outputs > Toolbox_run_output > Clustering

The clustering.py script performs a k-means clustering analysis to synthesize the information contained in the four habitat metrics. The objective is to group patches with similar multimetric profiles, thereby summarizing complex combinations of macroinvertebrate habitat quality and ecological risk into a smaller number of interpretable groups.

Because each metric represents a different dimension of macroinvertebrate habitat dynamics, clustering can reveal patterns that may not be apparent when the metrics are examined individually.

The optimal number of clusters is determined using the silhouette score. The maximum number of clusters considered is set to 9 by default and can be modified in the code if needed.

The clustering functions provide information on cluster statistics and Principal Component Analysis (PCA) loadings. The cluster number assigned to each patch is also added to the CSV and shapefile containing the metric results. Further information on the output files is provided in Section 5.3.

5. Output interpretation

Output location: data > output.

Relevant subfolders are used to organize the different types of outputs. The video 5_HaDy_output_analysis provides additional guidance on how to explore and interpret the outputs.

5.1 Outputs from the main.py file execution

Running *main.py* generates the outputs presented in Table 10.

Table 10 Outputs generated by main.py

Output name	Output location	Explanation
discharge_with_match.csv	Discharge_with_match	Associates each discharge in the input discharge time series with the closest discharge for which hydrodynamic simulation data are available.
cropped_discharge_and_wetted_area (csv and shapefile)	Cropped_shapefiles	Contains the input data cropped to the discharges considered in the analysis and to the spatial area of interest, corresponding to patches that are wet under the maximum discharge considered. This reduces the study domain before metric calculation and therefore reduces computational time.
habitat_classification.csv	Habitat_classification	Provides a static representation of the habitat classification for each patch and each discharge considered, before temporal integration. This file is used by the plotting functions to represent habitat availability at individual discharges.
metrics (csv and shapefile)	Metric_files	Contains the final metric results for patches that experience the target habitat type at least once during the habitat time series.
metrics_essentials.shp	Metric_files	Contains a selection of essential fields: patch ID, x-coordinate, y-coordinate, area, and the selected metrics.
Preview maps	Metric_files > Preview_maps	Provides a visual overview of the results before opening the GIS project. A preview map is generated for each selected metric.

Example case study Box 3: Outputs from the main.py file execution

The outputs from the example case study are located in data > output > Example_output.
The folder contains the files listed in Table 10, organized into their respective subfolders.

The extent of the zoomed area shown in the preview maps can be adapted by modifying the corresponding settings in crop_metric_and_map_preview.py. This allows the preview maps to be customized to focus on a specific area of interest.

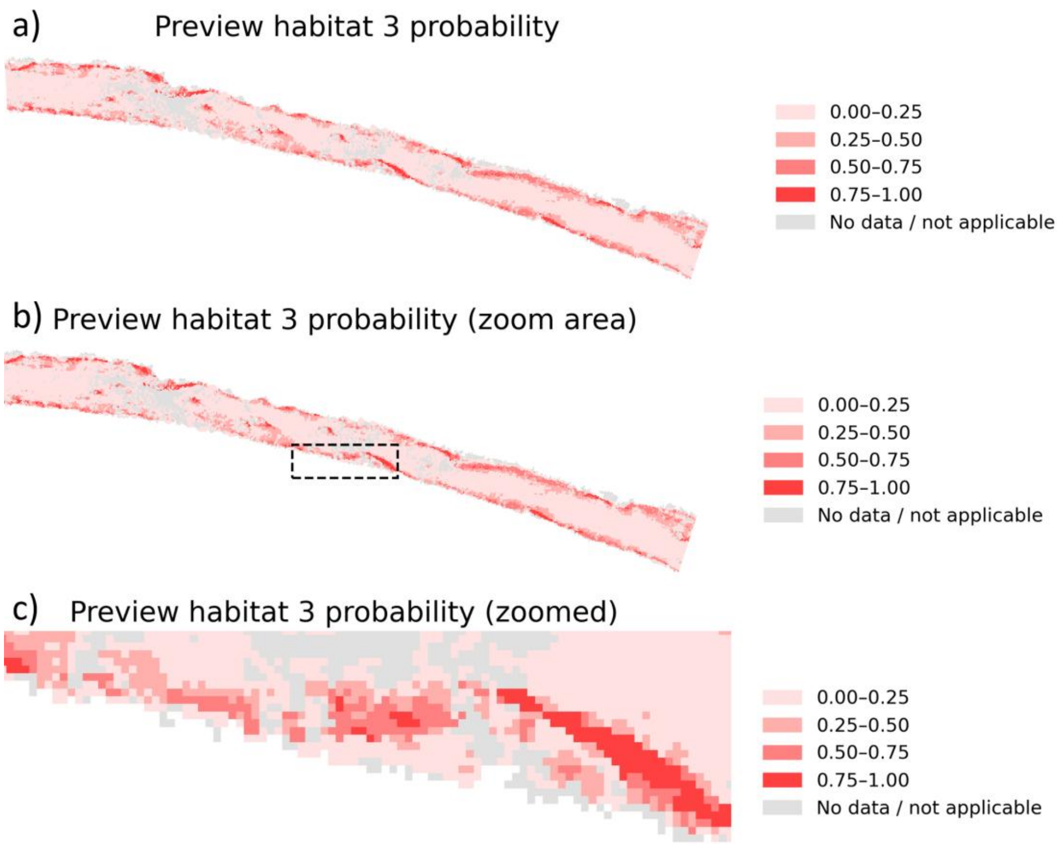


Figure 10 Preview maps from the example case study. Panel a) shows the full reach, panel b) identifies the zoomed area, and panel c) shows the corresponding zoomed area

For the example case study, both QGIS and ArcGIS projects are provided to facilitate exploration of the metric outputs. These are located in data > output > Example_output > QGIS_project.

For QGIS users, an example map template (example_case_maps_template.qpt) and default symbologies for the four metrics (.qml files) are available in the same folder.

For further guidance on exploring the results in QGIS, see the support video [3_HaDy_example_case_QGIS].

ATTENTION

Shapefile field names are limited to 10 characters. As a result, some field names in the output shapefile are shortened. Table 11 provides explanation.

In Table 11, "h" stands for habitat, and is followed by the corresponding habitat type number. As previously mentioned, metrics are calculated only for patches that experience the target habitat type at least once during the habitat time series.

Table 11 Description of the fields contained in the metric output files (CSV and shapefile)

Shapefile column name	Associated explanation
Id	Patch ID, identical to the corresponding field in the input shapefile.
x	X-coordinate of the patch, identical to the corresponding field in the input shapefile.
y	Y-coordinate of the patch, identical to the corresponding field in the input shapefile.
area	Patch area, identical to the corresponding field in the input shapefile.
d_discharge	Water depth for each discharge considered in the analysis, identical to the corresponding fields in the input shapefile.
v_discharge	Current velocity for each discharge considered in the analysis, identical to the corresponding fields in the input shapefile.
focus_zone	Only generated when focus_on_zone is true. Indicates whether the patch is located within the zone of interest: 0 if outside and 1 if inside.
id_focus	Only generated when focus_on_zone is true. Indicates the ID of the corresponding polygon in the zone-of-interest shapefile (e.g., the ID of an individual gravel bank).
Habitat probability	
dur_h	Duration of each habitat type during the habitat time series, expressed as a number of timesteps. The number of dur_h fields depends on the habitat classification. For example, the egg desiccation classification contains habitat types from -1 to 3, while the example community-level classification contains habitat types from 0 to 6.
prob_h	Probability of experiencing the corresponding habitat type, calculated as the duration of that habitat type divided by the total duration of the habitat time series.
mostProb	Most probable habitat type for each patch over the habitat time series, corresponding to the habitat type with the highest probability. This is calculated for all patches in the simulated domain, not only those experiencing the target habitat type (see Figure 10)
h_first	Timestep of the first occurrence of the corresponding habitat type. For example, h2_first gives the timestep of the first occurrence of habitat type 2 for a given patch.

Statistics on the target habitat type – advanced analysis focusing on a specific zone only	
h_nb_seq	Number of separate sequences during which the target habitat type occurs in the patch over the habitat time series.
h_dur_max	Maximum consecutive duration of the target habitat type in the patch, expressed in hours.
h_dur_med	Median consecutive duration of the target habitat type in the patch, expressed in hours.
h_dur_q1	First quartile of the distribution of consecutive durations of the target habitat type, expressed in hours.
h_dur_q3	Third quartile of the distribution of consecutive durations of the target habitat type, expressed in hours.
Habitat shifts	
h_sh_all	Total number of daily habitat shifts experienced by the patch over the habitat time series.
h_sh_suit	If not focusing on a zone: total number of daily shifts to the suitable habitat types specified in the configuration file. If focusing on a zone: total number of daily shifts to the target habitat type.
h_sh_dry	Total number of daily shifts to dry conditions over the habitat time series.
Desiccation	
h_dryMax	Maximum consecutive duration of dry conditions, expressed as a number of timesteps. The starting point of the calculation depends on start_at_first_occurrence: if false, the calculation starts at the beginning of the habitat time series; if true, it starts after the first occurrence of the target habitat type.
h_desicR	Associated desiccation risk value.
Desiccation additional statistics – advanced analyses focusing on a specific zone only	
h_dry_seq	Number of separate sequences of dry conditions in the patch over the habitat time series.
h_dry_max	Maximum consecutive duration of dry conditions in the patch, expressed in hours.
h_dry_med	Median consecutive duration of dry conditions in the patch, expressed in hours.
h_dry_q1	First quartile of the distribution of consecutive dry durations, expressed in hours.
h_dry_q3	Third quartile of the distribution of consecutive dry durations, expressed in hours.
h_dryMedR	Risk value associated with the median consecutive duration of dry conditions.
h_dry_q1R	Risk value associated with the first quartile of the distribution of consecutive dry durations.

h_dry_q3R	Risk value associated with the third quartile of the distribution of consecutive dry durations.
Drift	
h_dd_Risk	Duration spent in each drift risk category, expressed as a number of timesteps.
h_driftP	Drift risk percentile.
h_driftM	Maximum drift risk value.

5.2 Outputs from the plots.py execution

The *plots.py* script generates different outputs depending on the parameters selected in the configuration file (see Section 4.6, Plots).

<u>Output location:</u> data > output > Toolbox_run_output > Plots
Additional guidance on interpreting the outputs is provided in the video 5_HaDy_output_analysis.

Example case study Box 4: Outputs from the plots.py file execution

Static representation of habitat distribution per discharge

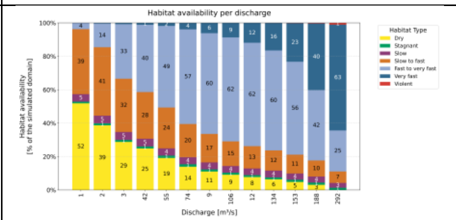
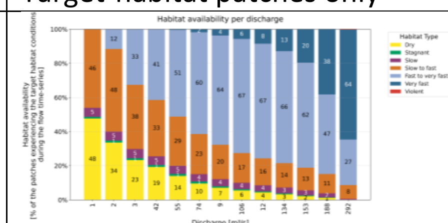
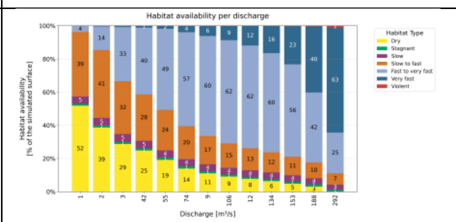
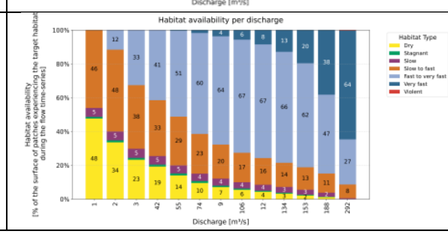
These figures show how habitat types are distributed across the study area at each simulated discharge.
The **x-axis** represents the different simulated discharges. For each discharge, the figures show the spatial distribution of habitat conditions within the selected spatial coverage.

Rows: Because patches do not all have the same area, habitat distributions are presented both:

- By **number of patches**, and
- By **total area** occupied by each habitat type.

Columns: Two spatial coverages are considered:

- **Entire simulated domain:** all patches that are wet under the maximum simulated discharge.
- **Target-habitat patches only:** only patches that experience the selected target habitat at least once during the habitat time series.

<table>
<tr><th></th><th>Entire simulated domain</th><th>Target-habitat patches only</th></tr>
<tr><td>Per number of patches</td><td>
</td><td>
</td></tr>
<tr><td>Per area</td><td>
</td><td>
</td></tr>
</table>

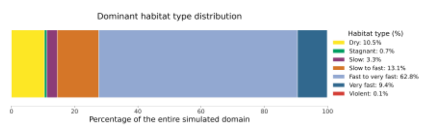
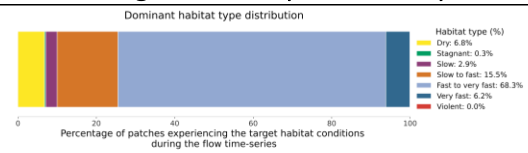
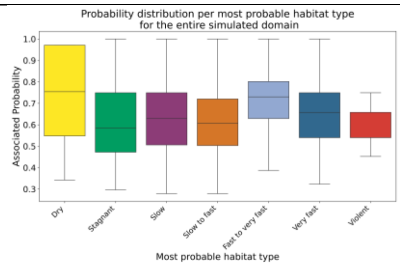
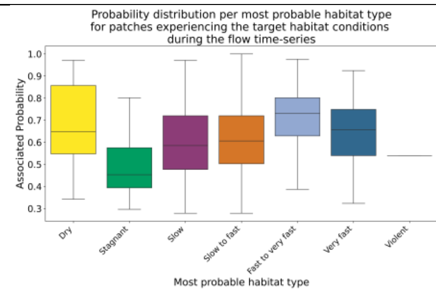
Most probable habitat type over the habitat time-series

For each patch, the **most probable habitat type** is the habitat type that occurs most frequently throughout the habitat time series.
The probability of each habitat type is calculated as:
Time spent in that habitat type ÷ Total duration of the habitat time series

These figures summarize the dominant habitat conditions by integrating habitat variability over both **space** and **time**.

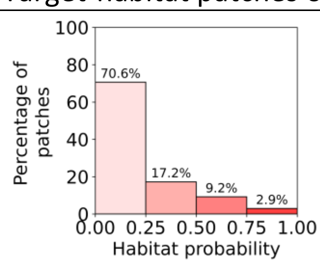
Columns: The two spatial coverages are identical to those described above:

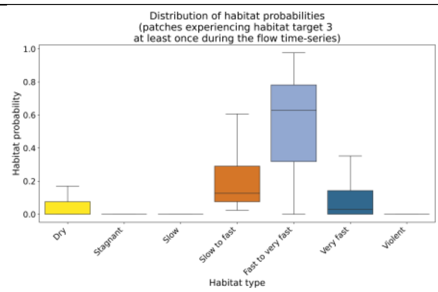
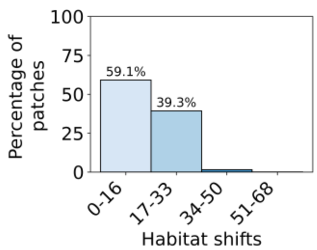
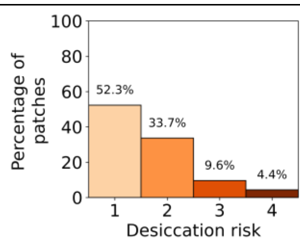
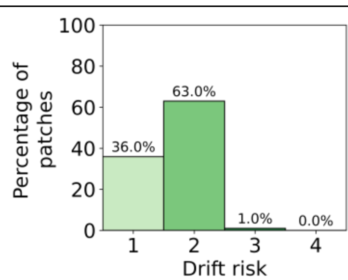
- The entire simulated domain, and
- Only patches that experience the target habitat during the habitat time series.

<table>
<tr><td></td><td>Entire simulated domain</td><td>Target-habitat patches only</td></tr>
<tr><td rowspan="2">Most probable habitat type distribution (number of patches)</td><td>
</td><td>
</td></tr>
<tr><td colspan="2">This figure shows which habitat type is the dominant one for each patch over the habitat time series.
For example, the habitat type "fast to very fast" is the most probable habitat type for 62.8% of the patches in the simulated study area, indicating that this habitat condition dominates most locations over time.</td></tr>
<tr><td rowspan="2">Probability associated with the most probable habitat type</td><td>
</td><td>
</td></tr>
<tr><td colspan="2">Knowing the dominant habitat type alone does not indicate how strongly that habitat dominates a patch. This figure therefore shows the probability associated with each patch's most probable habitat type.

For example, although "fast to very fast" is the dominant habitat for many patches, its associated median probability is approximately 0.75. This means that these patches spend about 75% of the habitat time series under "fast to very fast" conditions.

In this example, such high probabilities are consistent with the hydropeaking discharge regime, indicating that a large proportion of the riverbed experiences fast-flowing conditions for most of the simulated period.</td></tr>
<tr><td colspan="3">Metrics representations

The following figures are computed only for patches that experience the target habitat at least once during the habitat time series (target-habitat patches).
See Section 4.4 details on the metrics calculation</td></tr>
<tr><td></td><td>Target-habitat patches only</td><td>Explanation</td></tr>
<tr><td>Habitat probability of the target habitat type</td><td>
</td><td>This shows the probability that each patch is in the target habitat over the habitat time-series.
In this example, the target habitat is "slow to fast." Although the selected patches all experience this habitat at least once, most spend only a limited amount of time in it: 87.8% of the patches spend less than 50% of</td></tr>
</table>

		the habitat time series under target habitat conditions.
Habitat probability distributions		This presents the probability distributions of **all habitat types** for the target-habitat patches. It complements the previous figure by showing how the remaining time is distributed among the other habitat types. In this example, the selected patches spend most of their time under **"fast to very fast"** conditions, confirming the dominance of this habitat observed in the previous analyses.
Daily shifts to suitable habitats		This shows the number of **daily transitions into suitable habitat conditions** over the habitat time series. Suitable habitat types are specified by the user in the configuration file. In this example, the suitable habitats are: • Stagnant • Slow • Slow to fast • Fast to very fast • Very fast A shift is counted each time a patch changes from an **unsuitable habitat** (dry or violent current) to any of the defined suitable habitat types.
Desiccation risk		See Section 4.4 for details on the risk calculation. This figure shows the proportion of patches falling into each desiccation risk category, providing an overview of the spatial distribution of desiccation risk within the selected patches.
Drift risk		See Section 4.4 for details on the risk calculation. This figure shows the proportion of patches falling into each drift risk category, providing an overview of the spatial distribution of desiccation risk within the selected patches.

The available plotting functions are described in Section 4.6. Users can freely modify the plotting script to adapt the figures to their specific needs, for example by customizing existing plots or adding or removing figures.

To prevent these modifications from being overwritten when updating the toolbox from GitHub, any customized version of plots.py should be saved under a different filename or in a separate location.

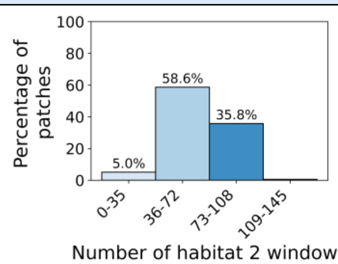
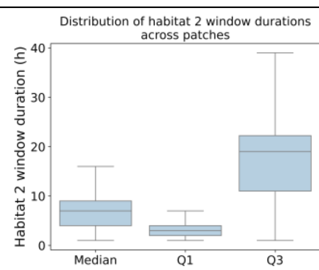
The generated figures depend on the parameters defined in the local.yaml configuration file. When patches have different sizes, some plots can be normalized by patch area instead of by number of patches, depending on the selected settings.

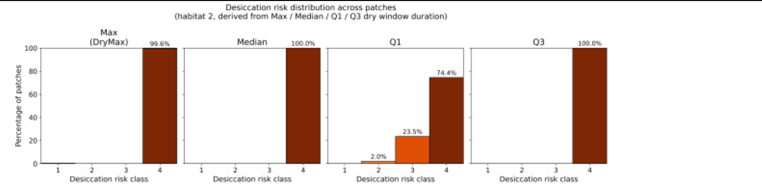
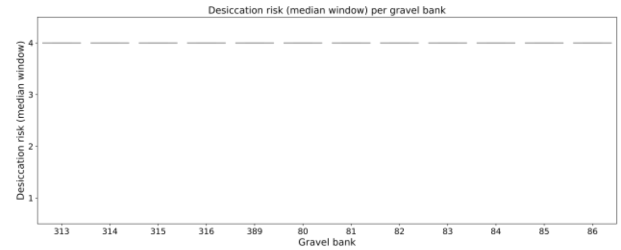
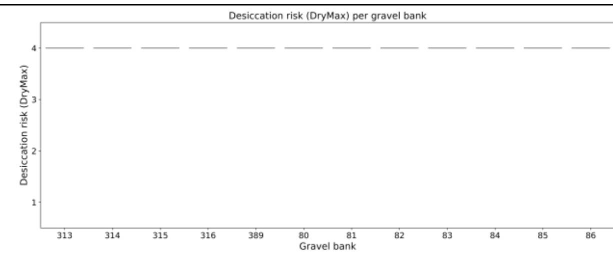
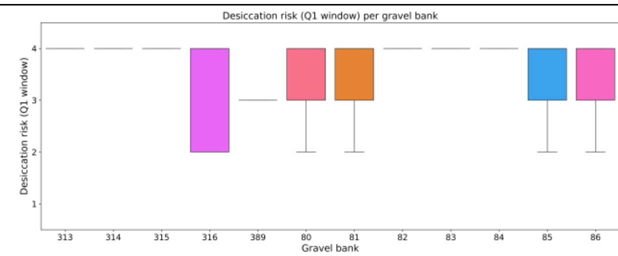
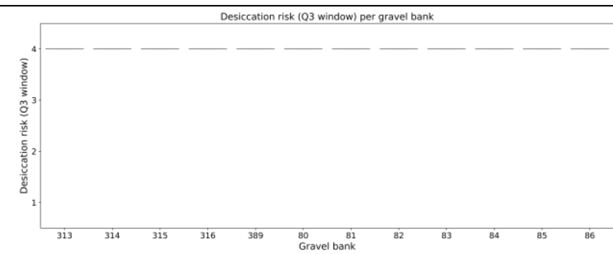
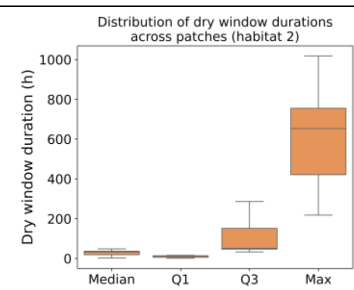
Advanced plots when focusing on a specific zone

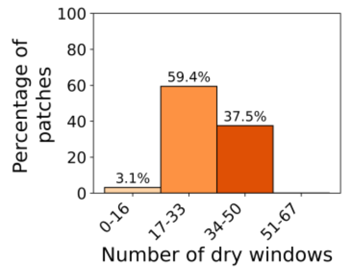
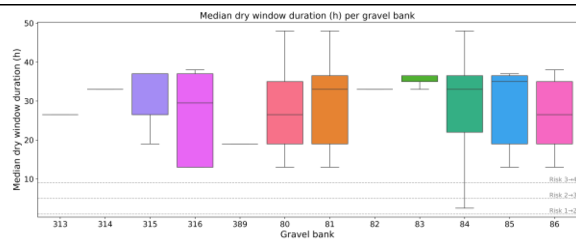
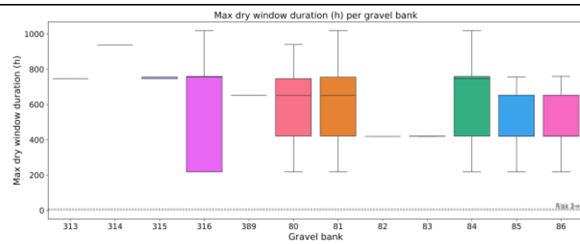
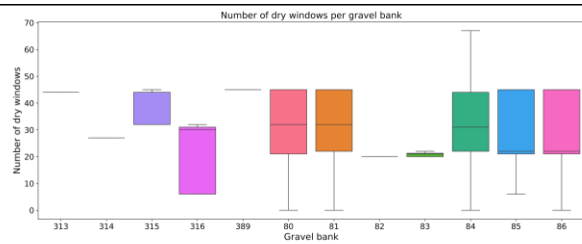
For more advanced analyses, additional plots are available when the focus_on_zone Boolean parameter is set to True. These outputs are presented in Table 12.

For example, when comparing different gravel banks, each patch must be assigned a zone ID indicating the gravel bank to which it belongs. The toolbox then generates plots separately for each zone, allowing direct comparison of habitat dynamics and associated metrics between gravel banks. Further information in the video 6_HaDy_advanced_focus_on_zone.

Table 12 Advanced plots generated by plot.py when focusing on a specific zone

Target habitat: additional statistics	
Number of individual sequences during which the target habitat occurs over the habitat time series.	
Distribution of the durations of target-habitat sequences, expressed in hours.	
Desiccation risk and dry durations	
Desiccation risk	

<table>
<tr><td>Distribution of desiccation risk values across all zones.</td><td>
</td></tr>
<tr><td>Median desiccation risk calculated for each gravel bank.</td><td>
</td></tr>
<tr><td>Desiccation risk values calculated for each gravel bank.</td><td>
</td></tr>
<tr><td>First quartile of desiccation risk values calculated for each gravel bank.</td><td>
</td></tr>
<tr><td>Third quartile of desiccation risk values calculated for each gravel bank.</td><td>
</td></tr>
<tr><td colspan="2">Dry durations</td></tr>
<tr><td>Distribution of dry window durations across all zones.</td><td>
</td></tr>
</table>

Number of separate dry periods identified across all zones.	
Median duration of dry periods for each gravel bank, expressed in hours.	
Maximum duration of dry periods for each gravel bank, expressed in hours.	
Number of separate dry periods identified for each gravel bank.	

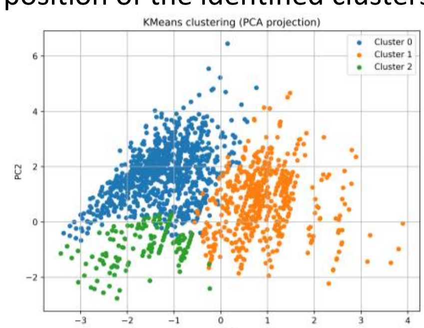
5.3 Outputs from the clustering.py execution

The *clustering.py* script generates different outputs depending on the parameters selected in the configuration file (see Section 4.6, Clustering).

<u>Output location:</u> data > output > Toolbox_run_output > Clustering

Additional guidance on interpreting the outputs is provided in the video [5_HaDy_output_analysis].

Example case study Box 5: Outputs from the clustering.py file execution

Output	Description
Hab_silhouette.png	Plot showing the silhouette score for different numbers of clusters. The number of clusters corresponding to the highest silhouette score is selected as the optimal number of clusters. 
Hab_PCA_clusters.png	Plot showing the position of the identified clusters in the  PC1–PC2 space.
Hab_PCA_loadings.csv	CSV file containing the PC1 and PC2 loadings for each metric.
metrics_with_clusters (csv and shapefile)	Metric output files supplemented with clustering results. Additional fields include Cluster_hab, indicating the cluster assigned to each patch, and PC1_hab and PC2_hab, containing the coordinates of each patch in the first two principal component dimensions.

6. Limitations and assumptions

The following limitations and assumptions should be considered when interpreting the toolbox outputs:

- **Input hydrodynamic data:** The toolbox assumes that the user has identified the discharges relevant to the study and performed the corresponding hydrodynamic simulations. The resulting water depth and current velocity data must be provided in the required input shapefile format.

- **Habitat classification and metric thresholds:** The toolbox provides example habitat classification thresholds and desiccation and drift risk thresholds. These values are not universally applicable and must be adapted to the objectives of the study, the habitat type considered, and the macroinvertebrate taxa of interest.

- **GIS analysis:** The toolbox provides output shapefiles and preview maps, but the detailed spatial exploration and interpretation of the results are performed by the user in a GIS software.

- **Discharge representation:** The toolbox assigns each discharge in the input time series to the closest discharge for which hydrodynamic simulation data are available. No interpolation is performed between simulated discharges. The representativeness and resolution of the selected simulated discharges therefore directly affect the resulting habitat time series and metrics.

- **Temporal resolution:** Metric calculations depend on the timestep of the input discharge time series. The timestep must accurately represent the temporal resolution required for the study.

- **Threshold-based risk assessment:** Drift and desiccation risks are estimated from predefined threshold values. These metrics should therefore be interpreted as relative or threshold-based indicators of exposure rather than direct measurements of demographic or population-level effects.

7. Trouble shooting

The following recommendations can help become familiar with the toolbox and avoid common issues:

- **Start small, in space and time**: Begin with a small spatial study area and a short discharge time series. This reduces computational time and makes it easier to check whether the results are consistent and plausible. To crop the original shapefile to a smaller study area, the QGIS procedure is described in Section 9.2
- **Save scripts and analysis files separately**: Pulling the latest version of the GitHub repository may overwrite locally modified scripts and files. Save any customized scripts, configuration files, and analysis results in a separate, safe location before updating the repository.
- **Follow the input formatting requirements carefully**: The format of the input data, particularly the column names in the input shapefile, is important because the toolbox extracts information from these names. For example, a column representing a discharge of 18.3 m^3/s must follow the required format, such as depth_18_3, rather than depth_18.3. If the required format is not respected, the toolbox may fail to run and no outputs will be generated. Column names can be modified using standard GIS software such as QGIS.
- **Check the local.yaml configuration file**: If no local.yaml file is created, the toolbox uses config.yaml by default, which contains the parameters of the example case study. For a tailored analysis, create and adapt local.yaml as described in Section 3.2, and save the file before running the toolbox.
- **Check that the configuration matches the input data**: Ensure that the parameters entered in local.yaml correspond to the actual input files. In particular, check the input filenames, column names and prefixes, discharge timestep, habitat classification, number of habitats, and selected metrics.
- **Test each modification progressively**: When adapting the toolbox, changing one parameter or function at a time makes it easier to identify the source of an issue. After each modification, save the relevant file before running the toolbox and check whether the resulting outputs are as expected.

8. Glossary

Patch: A spatially fixed unit within a river reach, typically corresponding to areas on the order of 0.25 to 100 m^2 (micro- to meso-scale). The appropriate patch size depends on the species and life stage considered, as well as their mobility. While the location of a patch remains constant over time, the habitat conditions within it may vary. In this toolbox, patches usually correspond to the mesh cells of the hydrodynamic simulation for which water depth and current velocity data are available.

Habitat: A set of environmental conditions characterized by dominant physical and chemical properties (e.g., current velocity, water depth, and substrate). These conditions can be described at different spatial scales, including the micro- to meso-scale within a river reach. Different species and life stages rely on specific habitat conditions to fulfil essential functions during their life cycle, such as feeding, reproduction, resting, or refuge seeking. The spatial distribution of habitats within a river reach can change over time.

Discharge time-series: The temporal evolution of river discharge (m^3/s), represented at a defined timestep (e.g., minutes or hours between two consecutive discharge values).

Habitat time-series: The temporal sequence of habitat conditions within each patch, obtained by combining the spatial habitat classification with the discharge time series.

Target habitat: The habitat type that is the primary focus of the analysis. Habitat types are defined according to a chosen habitat classification scheme, with each type assigned a numerical code for processing within the toolbox. Metrics are calculated specifically for patches that experience the target habitat at least once during the habitat time series.

Suitable habitats: Habitat types considered favorable for the macroinvertebrate taxa under study, based on the criteria defined in the selected habitat classification scheme.

Habitat probability (M1): A patch-scale metric representing the proportion of the habitat time-series during which a given habitat type occurs within a patch. It quantifies the temporal availability of habitat conditions within the patch and can be used to identify dominant habitat types.

Habitat shifts (M2): A patch-scale metric representing the number of daily transitions between habitat conditions within a patch. Depending on the selected configuration, shifts can be calculated between all habitat types, between suitable habitat types, or towards dry conditions. The metric provides an indication of habitat stability and temporal persistence.

Drift risk (M3): A patch-scale metric representing exposure to hydraulically induced drift. It is calculated from the drift-risk values obtained for the individual timesteps of the habitat time-series. The default metric is the 90^{th} percentile of these values, thereby representing frequent exposure to elevated drift conditions while reducing the influence of isolated extreme events.

Desiccation risk (M4): A patch-scale metric representing exposure to drying conditions over time. It is derived from the habitat time-series and reflects the occurrence and persistence of conditions in which the patch becomes dry due to insufficient water depth, indicating potential ecological stress for macroinvertebrates.

9. QGIS tips and tricks

9.1 Analyzing a shapefile value field on QGIS

The output shapefiles contain several attribute fields, corresponding to patch characteristics and calculated habitat metrics. An attribute field is a column in the shapefile attribute table containing a specific variable or metric for each patch.

To visualize a specific metric spatially, the corresponding attribute field can be selected as the basis for the layer symbology. Figure 11 provides an example of how to do this in QGIS.

For the example case study, a QGIS project and a map template are provided. The QGIS project can be opened directly to explore the example outputs with the predefined layers and symbologies.

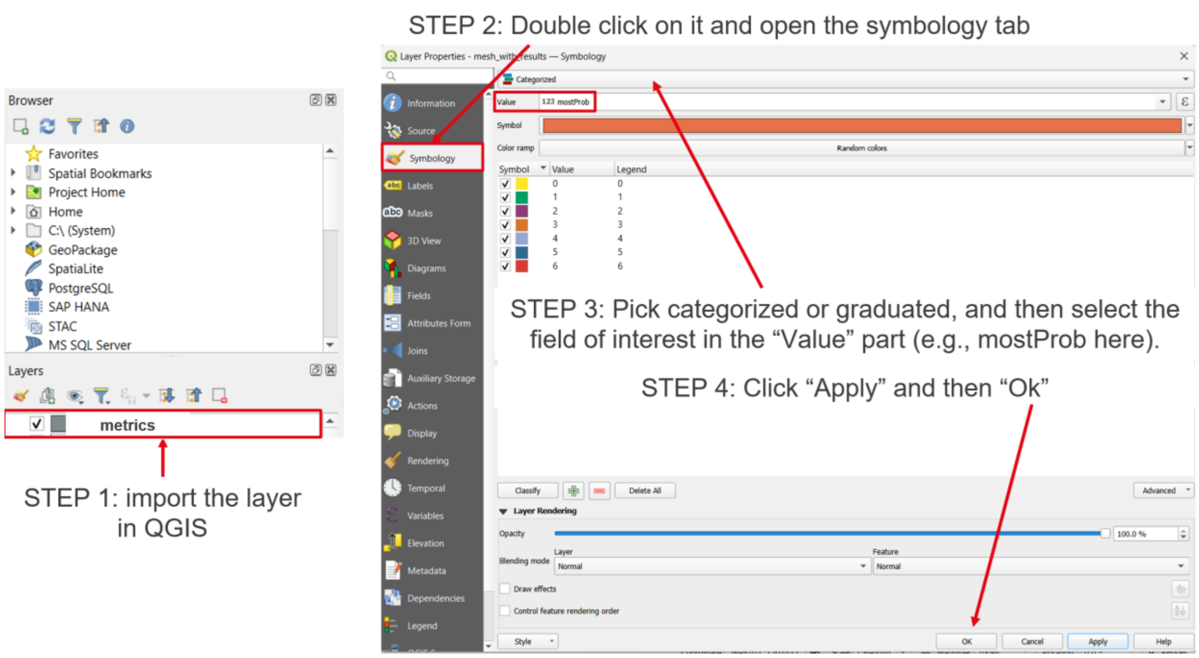


Figure 11 Example of how to visualize an output metric in QGIS

Default symbologies for the four metrics are also provided in: data > output > Example_output > GIS_project

The .qml files can be loaded in QGIS to apply the predefined symbology to the corresponding metric as follows:

1. Open the output shapefile in QGIS.
2. In the Layers panel, right-click the layer and select Properties.
3. Open the Symbology tab.

4. Select Style → Load Style.
5. Select the relevant .qml file from data > output > Example_output > GIS_project.
6. Click Apply and OK.

The example map template (.qpt) can be used to reproduce the layout of the example maps with other output layers. The template can be opened from QGIS using Project → New Print Layout from Template and then adapted to the desired study area, layers, and metrics.

9.2 Crop a shapefile to a specific area in QGIS

Cropping the input shapefile allows the creation of a reduced shapefile containing only the data within the area of interest. This can be useful when testing the toolbox on a smaller spatial extent, thereby reducing computational time.

Step 1. Prepare the layers

Two layers are required:

- the input shapefile to be cropped;
- a polygon mask defining the area of interest.
-

Step 2. Use the "Clip" tool

1. Open the Processing Toolbox from Processing → Toolbox.
2. Search for Clip.
3. Select Vector overlay → Clip.
4. In the dialog:
   - Input layer: select the shapefile to be cropped;
   - Overlay layer: select the polygon defining the area of interest;
   - Clipped: choose the location and name of the new shapefile.
5. Click Run.

The resulting shapefile contains only the features located within the mask polygon.

Step 3. Features with missing attribute values

The Clip tool can process features even when some attribute fields contain missing values. Missing attribute values therefore do not prevent the cropping operation.

However, missing values may affect subsequent analyses. If the resulting shapefile is subsequently used for statistical analyses, joins, or toolbox processing, the relevant fields should be checked and missing values handled according to the requirements described in Section 3.1.

Step 4. Optional: use "Extract by location"

If only features fully contained within the mask polygon should be retained, Extract by location can be used instead:

1. Open Vector → Research Tools → Extract by Location.
2. Select the input shapefile as the input layer.
3. Select the polygon mask as the comparison layer.
4. Select the within spatial relationship.
5. Run the tool.

This avoids retaining features that only partially overlap the area of interest.

9.3 Joining layers on QGIS

This procedure can be used when two input shapefiles need to be combined into a single shapefile. For example, one shapefile may contain the mesh cells with water depth and current velocity attributes, while another contains only the polygon geometry of the patches.

The desired output is a single polygon shapefile containing both the patch geometry and the associated attributes.

The two shapefiles should have a common identifier that can be used to establish the correspondence between features. If no common identifier is available, the layers can instead be joined based on their spatial location, provided that the features occupy corresponding locations.

Joining based on a common identifier

If both shapefiles contain the same unique patch identifier:

1. Open both shapefiles in QGIS.
2. Open the Processing Toolbox.
3. Search for Join attributes by field value.
4. Select the shapefile containing the patch geometry as the target layer.
5. Select the corresponding shapefile as the layer to join.
6. Select the common identifier field in both layers.
7. Select the output location and run the tool.

Joining based on spatial location

If the two shapefiles contain corresponding features at the same locations:

1. Select both layers in QGIS.

2. Create a spatial index for each layer if required:
3. Layer → Source → Geometry → Create Spatial Index.
4. Open the Processing Toolbox.
5. Search for Join attributes by location.
6. Select the polygon shapefile as the target layer.
7. Select the layer containing the attributes to be joined.
8. Select the appropriate spatial relationship.
9. Run the tool.

The resulting polygon shapefile contains the geometry of the target layer together with the attributes from the second shapefile.

10. Appendix

10.1 Technical setup support: prerequisites for using the toolbox

Installation of the core tools

The following tools are only recommendations. If you are used to working with python, conda virtual environments and GitHub, skip this part and use your usual tools.

Step 1: Download and install python

Get the latest version (minimum Python 3) from the official website: Download Python | Python.org, and follow the instructions.

Windows	macOS	Linux (Ubuntu/Debian)
Python Releases for Windows | Python.org Download install manager Check "Add Python to PATH"	Python Releases for macOS | Python.org	Python Source Releases | Python.org How to Install Python on Linux - GeeksforGeeks

Step 2: Install Git

From the official website: Git, and follow the instructions.

Windows	macOS	Linux (Ubuntu/Debian)
Git - Install for Windows Keep all default options	Git - Install for macOS	Git - Install for Linux

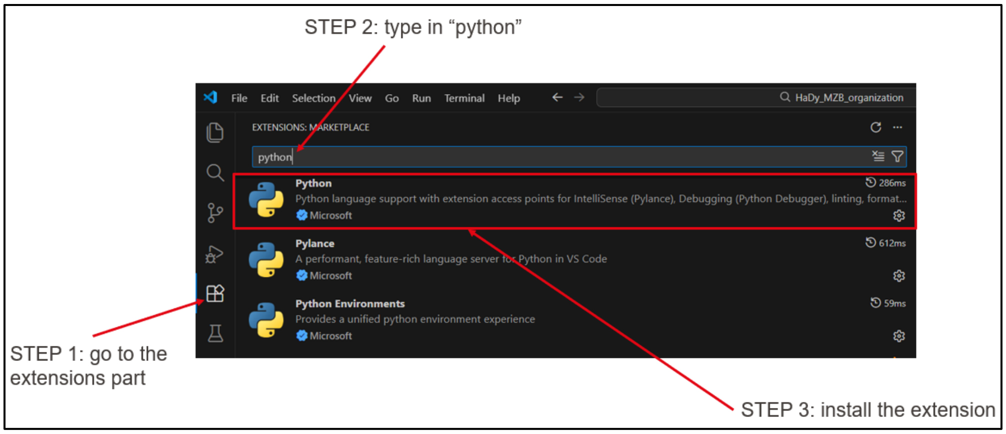
Step 3: Install anaconda navigator

From the official software: Installing Navigator - Anaconda, and follow the instructions.

Step 4: Install Visual Studio code and its python extension

From the official website: Download Visual Studio Code - Mac, Linux, Windows

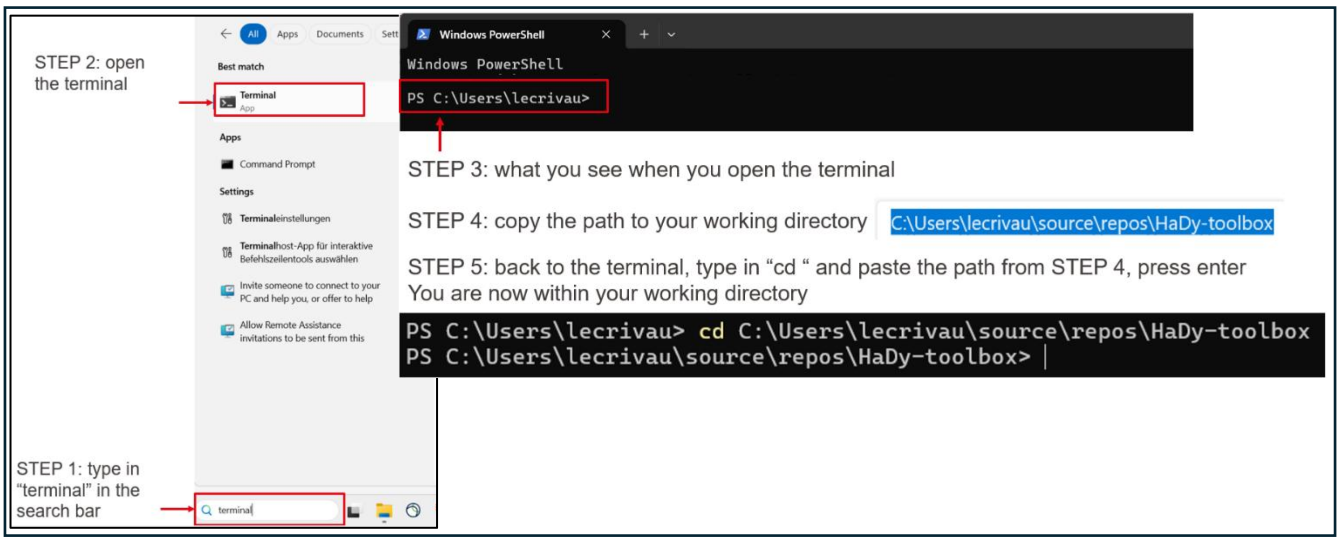
For Microsoft users only, install the python extension: open VS Code and follow instructions.



Basic commands: accessing a folder on a terminal

To open a terminal, open the search bar, type in "terminal", and open it (see below for Windows).

Then type in the command cd and then your folder absolute path.



10.2 Description of files and functions

Main.py

The **main.py** file collects functions from other python files and variables entered by the user in the configuration file and executes the code.

Description of the workflow (in chronological order)	Name of the function	Where the function is located	Name of the output
Getting the known discharge values from the column names of	get_discharge_values	*support_functions.py*	discharge_values

the input polygon shapefile			
Match the discharge of the discharge time-series to the closest known one in discharge_values	match_closest_discharge	*discharge_time_series.py*	discharge_with_match.csv
Select relevant discharges for the analysis: the list of corresponding known discharges	get_study_discharges	*discharge_time_series.py*	relevant_discharges
Crop the shapefile: to the corresponding relevant discharges; and keeping only the wetted mask (patches wet for the max considered discharge)	prepare_wetted_shapefile_for_relevant_discharges	*support_functions.py*	cropped_discharge_and_wetted_area.shp
Export the cropped shapefile to a csv	prepare_csv	*habitat_classification.py*	cropped_discharge_and_wetted_area.csv
If focus_on_zone is true			
Attribute habitat only for patches in the focus zone (otherwise -1 by default). This habitat classification function needs to be adapted by the user	attribute_habitat_types_zone_only	*habitat_attribution.py*	habitat_classification.csv
Metrics calculation	process_mesh_data_focus_on_zone	*metrics_calculation_focus_on_zone.py*	metrics.csv
Save the metrics to a shapefile	join_mesh_with_CSV_data	*support_functions.py*	metrics.shp
If focus_on_zone is false			
Attribute habitats based on the example habitat classification	attribute_habitat_current_based	*habitat_classification.py*	habitat_classification.csv
Metrics calculation	process_mesh_data	*metrics_calculation.py*	metrics.csv
Save the metrics to a shapefile	join_mesh_with_CSV_data	*support_functions.py*	metrics.shp

Crop_metric_and_map_preview.py

The *crop_metric_and_map_preview.py* script provides a quick visual overview of the metric results by cropping the metric shapefile to a user-defined area and generating preview maps. It is intended to facilitate the initial exploration of the spatial distribution of the calculated metrics before opening the complete dataset in a GIS software.

The zoomed area can be defined manually by entering the coordinates of the desired extent directly in the relevant function in the script. This allows the user to select a specific area of interest independently of the automatically generated preview extent. The coordinates can be adapted according to the study area and the spatial feature to be examined.

The script can be modified by the user to change the selected area, the metrics displayed, or the appearance of the preview maps.

Description of the workflow (in chronological order)	**Name of the function**	**Name of the output**
Crop the metric shapefile to a specific area of interest	*crop_metric_shapefile*	Cropped metric shapefile
Generate preview maps for the selected metrics	*map_preview*	Preview maps of the selected metrics

Plot.py

The ***plot.py*** contains the function for the different plots. The user can always add and/or modify these functions (more information in Section 4.6 and Section 5.2).

Description of the workflow (in chronological order)	**Name of the function**	**Name of the output**
ALL PATCHES IN SIMULATED DOMAIN Plot most probable habitat type and related intensity	*plot_most_prob_intensity*	hist_most_prob_and_associated_intensity_simulated_zone.png
TARGET HABITAT Plot most probable habitat type and related intensity	*plot_most_prob_intensity_target_hab*	hist_most_prob_and_associated_intensity_target_habitat.png
ALL PATCHES IN SIMULATED DOMAIN Percentage of most probable habitat types	*plot_most_prob_percentages*	most_prob_percentages_simulated_zone.png
TARGET HABITAT Percentage of most probable habitat types	*plot_most_prob_percentages_target_hab*	most_prob_percentages_target_hab.png

ALL PATCHES IN SIMULATED DOMAIN Horizontal stacked bar of dominant habitat types	*plot_most_prob_horizontal*	most_prob_horizontal_simulated_zone.png
TARGET HABITAT Horizontal stacked bar of dominant habitat types	*plot_most_prob_horizontal_target_hab*	most_prob_horizontal_target_hab.png
ALL PATCHES IN SIMULATED DOMAIN IN PERCENTAGE OF PATCHES - Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge*	habitat_availability_per_discharge_simulated_domain.png
TARGET HABITAT IN PERCENTAGE OF PATCHES - Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge_target_hab*	habitat_availability_per_discharge_target_hab.png
FUNCTION: ALL PATCHES IN SIMULATED DOMAIN IN AREA – Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge_area*	habitat_availability_per_discharge_simulated_domain_area.png
TARGET HABITAT IN AREA – Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge_area_target_hab*	habitat_availability_per_discharge_simulated_domain_target_hab_area.png
TARGET HABITAT Habitat probability for each habitat type (Among all patches that experience the target habitat, what is the distribution of time spent in each habitat?)	*plot_probability_distribution_target_hab*	habitat_probabilities_target_hab.png
TARGET HABITAT Plot percentage histograms for selected metrics of a target habitat	*plot_histograms_target_habitat*	histogram_prob_percentage.png histogram_shifts_percentages.png histogram_desicc_percentages.png histogram_drift_percentages.png boxplot_dry_window_duration.png

		boxplot_targ_window_duration.png
TARGET HABITAT Compare desiccation metrics across gravel banks (id_focus) using Kruskal-Wallis and Dunn post-hoc tests, with boxplots per polygon.	*plot_desiccation_by_polygon*	desicc_max_h_by_polygon.png desicc_median_h_by_polygon.png desicc_risk_by_polygon.png desicc_risk_median_by_polygon.png desicc_risk_q1_by_polygon.png desicc_risk_q3_by_polygon.png desicc_window_count_by_polygon.png desiccation_summary_statistics.csv

Clustering.py

The ***clustering.py*** contains the function regarding the clustering presented in Section 4.6 and Section 5.3. The user can always add and/or modify this file.

Description of the workflow (in chronological order)	Name of the function	Name of the output
Performing the clustering: Finding the best number of clusters, PCA analysis, cluster statistics	*perform_clustering_target_habitat*	metrics_with_clusters.csv PCA_clusters.png PCA_loadings.csv silhouette.png
Join a GeoDataFrame (patch geometry) with a results DataFrame and save to a new shapefile	*join_mesh_with_CSV_data*	metrics_with_clusters.shp

11. Literature references

Bätz N, Judes C, Vanzo D, et al (2025) Patch-scale habitat dynamics: three metrics to assess ecological impacts of frequent hydropeaking. Journal of Ecohydraulics 10:79 – 106. https://doi.org/10.1080/24705357.2024.2426790

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